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Reliability Value-added Convenience

Longi Magnet Co., Ltd.

The Most Competitive Supplier of Industrial
Magnetic Equipment Globally



Tramp Magnet

> Content

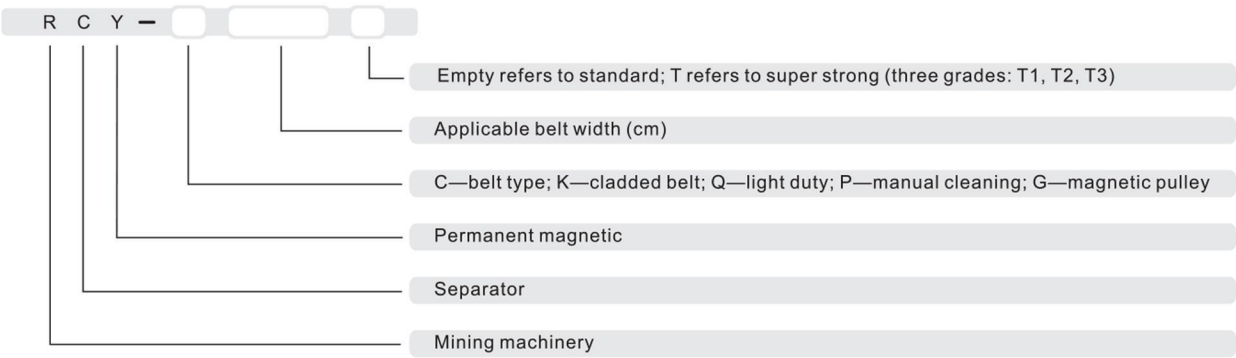
RCY-C Belt Permanent Magnetic Separator	5
RCYK-C Cladded Belt Permanent Magnetic Separator	6
RCY-G Magnetic Pulley	7
RCY-P Manual-cleaning Permanent Magnetic Separator	9
RCY-Q Light-duty Self-cleaning Permanent Magnetic Separator	10
ZWCY Manual-cleaning Permanent Magnetic Separator	11
RCDA Fan-cooled Manual-cleaning Electromagnetic Separator	12
RCDB Air-cooled Manual-cleaning Electromagnetic Separator	13
RCDC Fan-cooled Self-cleaning Electromagnetic Separator	14
RCDD Air-cooled Self-cleaning Electromagnetic Separator	15
RCDE/F Oil-cooled Electromagnetic Separator	16
RCDK Cladded Belt Electromagnetic Separator	17
PDC Manual-cleaning Electromagnetic Separator	18
LES Eddy Current Separator	19

> Instruction

Features and Application Of Permanent Magnetic Separator

Permanent magnetic separators are applied to remove tramp iron over various belt conveyor or chutes, and features with compact structure, free of cooling system and excitation system, and automatic rectification of departure of discharge belt, which are suitable for harsh working environment.

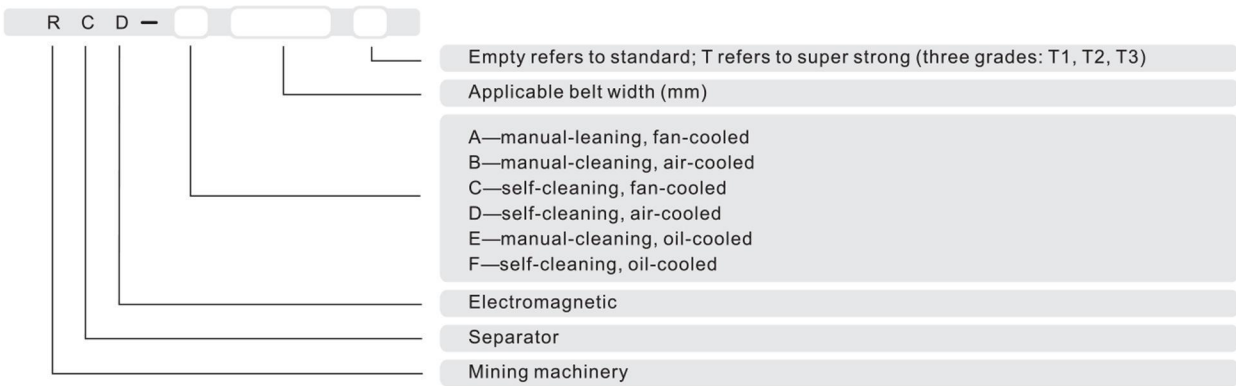
Model Instruction



Features and Application Of Electromagnetic Separator

Electromagnetic separators are applied to remove tramp iron over various belt conveyor or chutes, which are widely applied in power station, mines, metallurgy plants, construction materials food, etc. They are featured as low temperature rise, good insulation, strong magnetic force, low power consumption and long-term reliable performance.

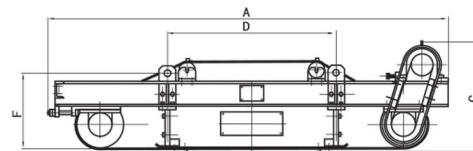
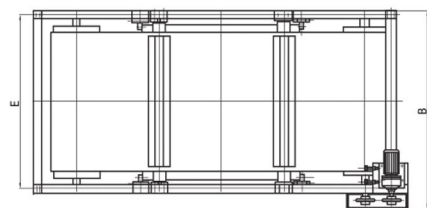
Model Instruction



> RCY-C Belt Permanent Magnetic Separator

Features

- Remove tramp iron buried in thick material layer from belt conveyor continuously.
- Quality rare earth magnet.
- Wide range of magnetic intensity available.
- Compact structure, low noise, easy maintenance, energy saving.
- Rectify departure of discharge belt automatically.



Technical Parameter

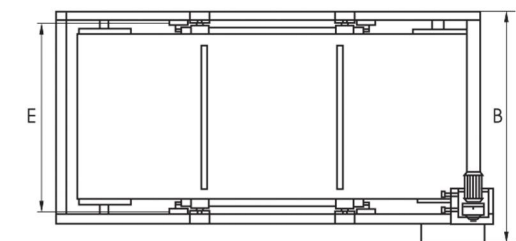
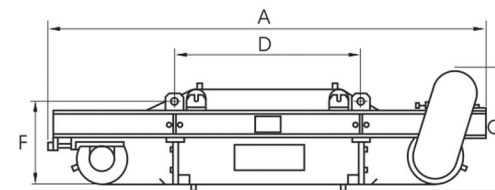
Model	Applicable Belt Width	Belt Speed	Rated Suspension Height	Magnetic Intensity at RSH	Catching Capacity	Duty Cycle	Motor Power	Overall Dimension mm						Weight
	mm	m/s	mm	mT	kg		kw	A	B	C	D	E	F	kg
RCY-C50	500	≤4.5	150	70	0.1-35	Continuously	1.5	1970	903	722	620	647	451	700
RCY-C65	650	≤4.5	200	70	0.1-35	Continuously	1.5	2110	1053	722	770	797	451	1000
RCY-C80	800	≤4.5	250	70	0.1-35	Continuously	2.2	2320	1253	722	970	997	451	1300
RCY-C100	1000	≤4.5	300	70	0.1-35	Continuously	3.0	2690	1453	803	1120	1190	451	1700
RCY-C120	1200	≤4.5	350	70	0.1-35	Continuously	4.0	2890	1713	868	1320	1430	516	2300
RCY-C140	1400	≤4.5	400	70	0.1-35	Continuously	4.0	3090	1913	868	1520	1630	516	3100
RCY-C160	1600	≤4.5	450	70	0.1-35	Continuously	5.5	3260	2115	868	1720	1830	516	4000
RCY-C180	1800	≤4.5	500	72	0.1-35	Continuously	7.5	2540	2440	977	1860	2120	516	5300
RCY-C200	2000	≤4.5	550	72	0.1-35	Continuously	7.5	3540	2635	1010	1860	2315	570	7700

We reserve the right to change the above specification without prior notice.

> RCYK-C Cladded Belt Permanent Magnetic Separator

Feature

- Custom designed for concrete recycling, construction, demolition, and waste processing industry.
- 1mm to 5mm thick stainless steel clad protects belt effectively.
- Strong magnetic force, high magnetic gradient.
- Compact structure, easy maintenance, automatic rectification of departure of discharge belt, long-term stable performance.



Technical Parameter

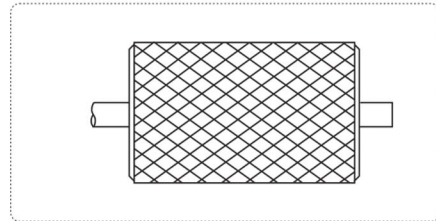
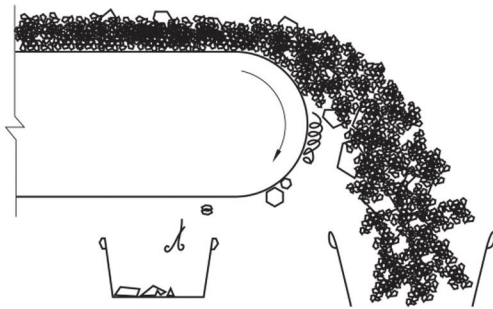
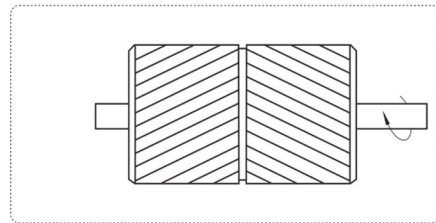
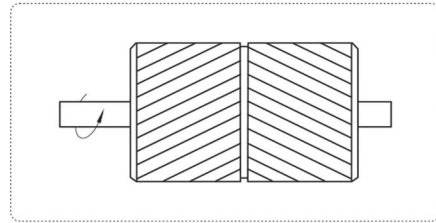
Model	Applicable Belt Width	Belt Speed	Rated Suspension Height	Magnetic Intensity at RSH	Catching Capacity	Duty Cycle	Motor Power	Overall Dimension mm						Weight
	mm	m/s	mm	mT	kg		kw	A	B	C	D	E	F	kg
RCYK-C50	500	≤4.5	150	70	0.1-35	Continuously	1.5	1970	903	722	620	647	451	700
RCYK-C65	650	≤4.5	200	70	0.1-35	Continuously	1.5	2110	1053	722	770	797	451	1150
RCYK-C80	800	≤4.5	250	70	0.1-35	Continuously	2.2	2320	1253	722	970	997	451	1380
RCYK-C100	1000	≤4.5	300	70	0.1-35	Continuously	3.0	2690	1453	803	1120	1190	451	1780
RCYK-C120	1200	≤4.5	350	70	0.1-35	Continuously	4.0	2890	1713	868	1320	1430	516	2390
RCYK-C140	1400	≤4.5	400	70	0.1-35	Continuously	4.0	3090	1913	868	1520	1630	516	3210

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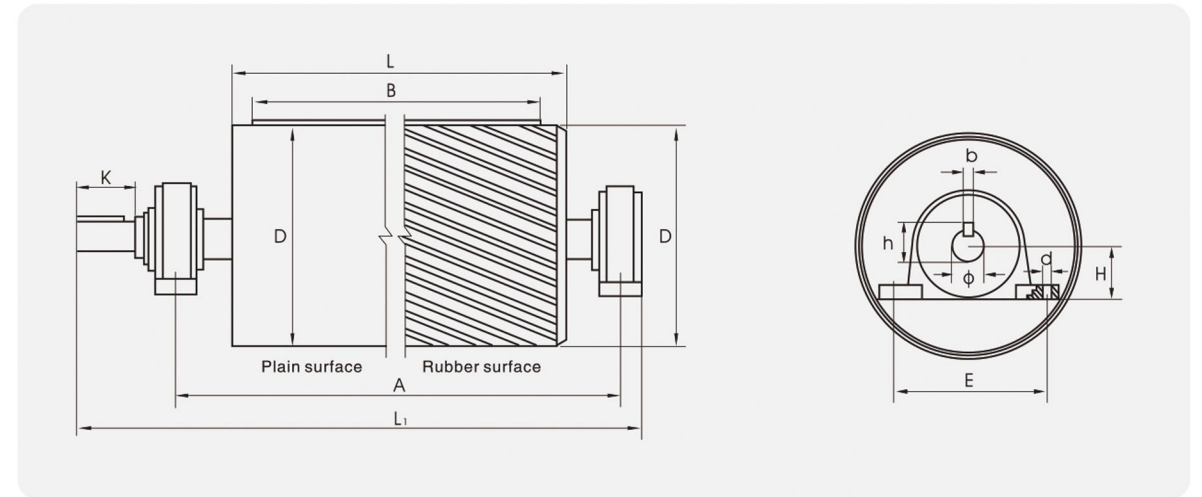
> RCY-G Magnetic Pulley

Feature

- Available in wide range: magnetic intensity from 1500Gs to 5000Gs, pulley diameter from 500mm to 1400mm.
- Compact structure, easy maintenance, reliable performance.
- Excellent removal of tramp iron buried deeply in material.



> RCY-G Magnetic Pulley



Technical Parameter

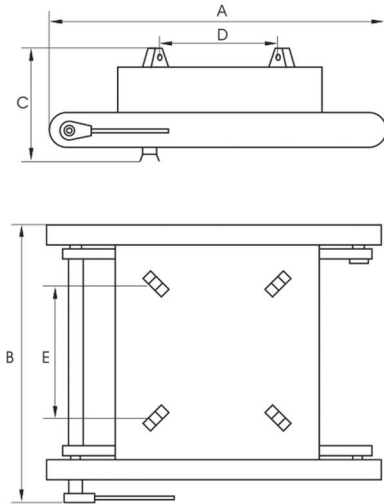
Model	Drum Length	Drum Diameter	Spindle Nose	Linchpin		Bearing Pedestal				Bearing Model	Total Length	Torsion KN·m		Weight
	L	D	Φ×K	b	h	A	E	H	d		L ₁	Plain surface	Rubber surface	kg
RCY-G50	600	500	55×115	16	60	850	240	80	27	1312	1097	1.63	2.5	380
RCY-G65-1	750	500	70×135	20	76	1000	350	120	27	1316	1280	2.12	3.26	580
RCY-G65-2	750	630	70×135	20	76	1000	350	120	27	1316	1280	2.96	4.54	800
RCY-G80-1	950	500	70×135	20	76	1300	350	120	27	1316	1580	2.62	4.01	765
RCY-G80-2	950	630	90×175	24	97	1300	380	140	27	3520	1661	3.67	5.61	1090
RCY-G80-3	950	800	90×175	24	97	1300	380	140	27	3520	1661	5.59	8.55	1480
RCY-G100-1	1150	630	90×175	24	97	1500	380	140	27	3520	1861	4.57	7.01	1285
RCY-G100-2	1150	800	110×215	32	119	1500	440	160	34	3524	1945	6.96	10.68	1865
RCY-G100-3	1150	1000	130×255	36	140	1500	480	180	34	3528	2020	—	17.75	2795
RCY-G120-1	1400	630	110×215	32	119	1750	440	160	34	3524	2195	5.49	8.4	1610
RCY-G120-2	1400	800	110×215	32	119	1750	440	160	34	3524	2195	8.37	12.81	2465
RCY-G120-3	1400	1000	130×255	36	140	1750	480	180	34	3528	2270	—	21.3	3560
RCY-G120-4	1400	1250	150×275	40	161	1750	540	200	34	3532	2305	—	30.0	5230
RCY-G140-1	1600	800	110×215	32	119	2000	440	160	34	3524	2445	9.76	14.92	2865
RCY-G140-2	1600	1000	150×275	40	161	2000	540	200	34	3532	2555	—	24.93	4405
RCY-G140-3	1600	1250	150×275	40	161	2000	540	200	34	3532	2555	—	35.0	6005
RCY-G140-4	1600	1400	170×335	40	181	2000	590	220	40	3536	2635	—	47.0	7535

Magnetic Intensity : Special T=300~400mT; Standard P=150~250mT.

> RCY-P Manual-cleaning Permanent Magnetic Separator

Feature

- Applied over narrow belt conveyor, thin material layer, small amount of tramp iron.
- Manual cleaning.
- Compact structure, easy maintenance, reliable performance, noise-free.



Technical Parameter

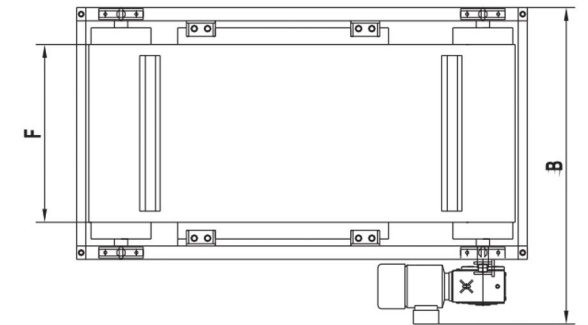
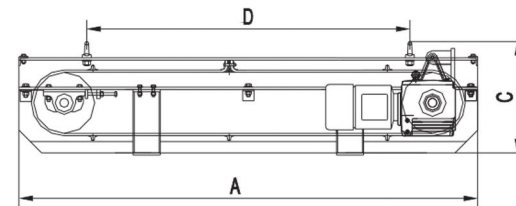
Model	Applicable Belt Width	Belt Speed	Rated Suspension Height	Magnetic Intensity at RSH	Catching Capacity	Suspension Size D×E	Overall Dimension A×B×C	Weight
	mm	m/s	mm	mT	kg	mm	mm	kg
RCY-P40	400	≤4.0	125	>70	0.1-30	340×340	900×850×400	310
RCY-P50	500	≤4.0	150	>70	0.1-30	460×460	1000×875×425	460
RCY-P65	650	≤4.0	200	>70	0.1-30	600×600	1150×1025×438	630
RCY-P80	800	≤4.0	250	>70	0.1-30	770×770	1350×1225×480	910
RCY-P100	1000	≤4.0	300	>70	0.1-30	1000×1000	1500×1375×510	1240

We reserve the right to change the above specification without prior notice.

> RCY-Q Light-duty Self-cleaning Permanent Magnetic Separator

Feature

- Easy maintenance, simple installation.
- Small volume, low noise, light weight, compact structure.
- Applied in environmental protection, food, molding, mining, etc., especially recycling industry.



Technical Parameter

Model	Applicable Belt Width	Rated Suspension Height	Motor Power	Overall Dimension mm						Weight
	mm	mm	kw	A	B	C	D	E	F	kg
RCY-Q60	600	250	2.2	1880	1425	520	900	1073	800	1150
RCY-Q80	800	250	2.2	2030	1425	520	1430	1073	800	1300
RCY-Q100	1000	250	2.2	2030	1610	520	1430	1073	800	1400
RCY-Q120	1200	250	2.2	2450	1425	520	1300	1073	800	1650
RCY-Q140	1400	250	4.0	2800	1450	520	2000	1073	800	1900

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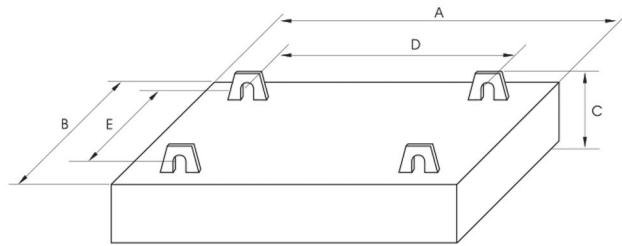
> ZWCY Manual-cleaning Permanent Magnetic Separator

Application

ZWCY series magnetic separator is applied for removal of tramp iron from various materials in the process of crushing, screening and conveying.

Feature

- Stable and reliable magnetic field
- Strong adaptability to harsh working environment except high temperature
- No power consumption
- Compact structure, light weight and easy maintenance



Technical Parameter

Model	Applicable Belt Width	Rated Suspension Height	Magnetic Intensity at RSH	Catching Capacity	Magnet Style	Overall Dimension mm					Weight
	mm	mm	mT	kg		A	B	C	D	E	kg
ZWCY-50A	500	120	50	0.1-25	Rare Earth	528	348	117	280	280	76
ZWCY-65A	650	150	50	0.1-25	Rare Earth	628	348	130	280	280	97
ZWCY-80A	800	200	50	0.1-25	Rare Earth	828	508	160	420	420	248
ZWCY-100A	1000	230	50	0.1-25	Rare Earth	1028	658	160	550	550	390
ZWCY-50B	500	100	45	0.1-25	Ferrite	598	488	209	400	400	165
ZWCY-50C	500	170	45	0.1-25	Ferrite	728	573	209	500	500	284
ZWCY-65B	650	120	45	0.1-25	Ferrite	728	573	191	500	500	257
ZWCY-65C	650	180	45	0.1-25	Ferrite	828	658	219	550	550	420
ZWCY-80B	800	180	45	0.1-25	Ferrite	828	658	219	550	550	420
ZWCY-100B	1000	200	45	0.1-25	Ferrite	1028	658	219	550	550	496

We reserve the right to change the above specification without prior notice.

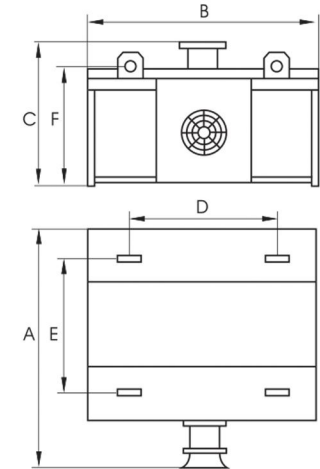
> RCDA Fan-cooled Manual-cleaning Electromagnetic Separator

Application

Removal of tramp iron in outdoor installation and light-dust working site.

Feature

- Coils are anti-oxidation, anti-rust and well-insulated.
- Large heat dissipation area ensures low temperature rise and excellent cooling performance.
- Wide range of magnetic intensity: 500Gs, 700 Gs, 1200 Gs, 1500Gs, etc.



Technical Parameter

Model	Applicable Belt Width	Cooling Style	Rated Suspension Height	Magnetic Intensity at RSH	Fan Power	Overall Dimension mm					Weight
	mm		mm	mT	kw	A	B	C	D	E	kg
RCDA-5	500	Fan cooled	150	70	0.37	1090	730	411	550	550	400
RCDA-6	650	Fan cooled	200	70	0.37	1160	800	512	600	600	650
RCDA-8	800	Fan cooled	250	70	0.37	1260	900	580	700	700	1020
RCDA-10	1000	Fan cooled	300	70	0.55	1360	1000	669	750	750	1600
RCDA-12	1200	Fan cooled	350	70	0.55	1560	1200	759	850	850	2300
RCDA-14	1400	Fan cooled	400	70	1.1	1760	1400	843	900	900	3400
RCDA-16	1600	Fan cooled	450	70	1.1	1960	1600	913	1000	1000	4500

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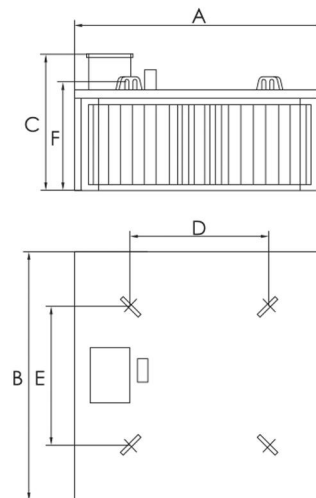
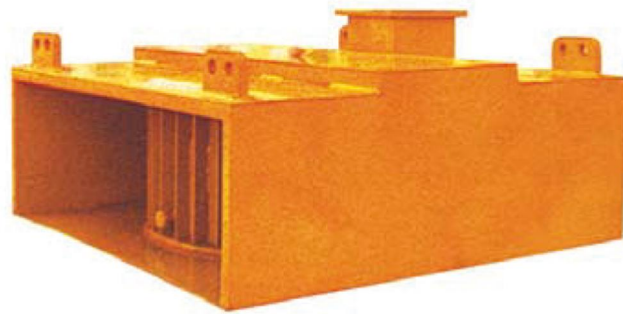
> RCDB Air-cooled Manual-cleaning Electromagnetic Separator

Application

Suitable for various working sites, especially bad working conditions.

Feature

- Compact structure, small volume, light weight.
- Maintenance free, noise free.
- Good heat dissipation, low temperature rise.
- Optional magnetic intensity at RSH: 500Gs, 700 Gs, 1200 Gs, 1500Gs, etc.



Technical Parameter

Model	Applicable Belt Width	Cooling Style	Rated Suspension Height	Magnetic Intensity at RSH	Excitation Power	Overall Dimension mm						Weight
	mm					A	B	C	D	E	F	
RCDB-5	500	Air cooled	150	70	1.8	920	820	431	550	550	353	600
RCDB-6	650	Air cooled	200	70	2.5	1020	920	519	600	600	441	1000
RCDB-8	800	Air cooled	250	70	3.4	1120	1130	599	700	700	526	1550
RCDB-10	1000	Air cooled	300	70	4.4	1250	1200	691	750	750	618	2200
RCDB-12	1200	Air cooled	350	70	7.6	1290	1300	771	850	850	698	2900
RCDB-14	1400	Air cooled	400	70	9.1	1400	1400	841	900	900	768	3800
RCDB-16	1600	Air cooled	450	70	11.0	1600	1600	933	1000	1000	860	5150

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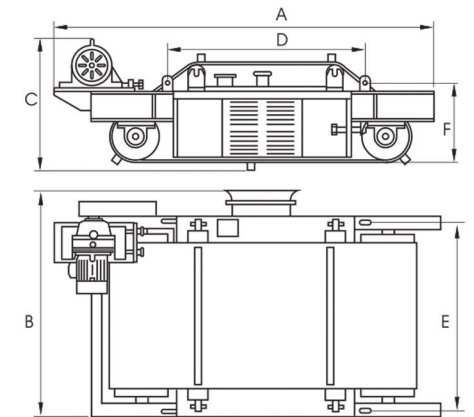
> RCDC Fan-cooled Self-cleaning Electromagnetic Separator

Application

RCDC Fan-cooled Self-cleaning Electromagnetic Separator

Feature

- Compact structure, easy maintenance.
- Automatic rectification of departure of discharge belt.
- Low temperature rise.
- Continuous removal of tramp iron.



Technical Parameter

Model	Applicable Belt Width	Cooling Style	Rated Suspension Height	Magnetic Intensity at RSH	Excitation Power	Motor Power	Fan Power	Overall Dimension mm						Weight
	mm							A	B	C	D	E	F	
RCDC-5	500	Fan cooled	150	70	2.0	1.5	0.37	2300	1143	732	746	723	461	730
RCDC-6	650	Fan cooled	200	70	3.0	1.5	0.37	2370	1286	732	816	873	461	1070
RCDC-8	800	Fan cooled	250	70	4.1	2.2	0.37	2550	1493	732	922	1073	461	1650
RCDC-10	1000	Fan cooled	300	70	5.9	3.0	0.55	2800	1693	800	1022	1273	484	2400
RCDC-12	1200	Fan cooled	350	70	7.0	4.0	0.55	3200	1953	878	1230	1523	595	3400
RCDC-14	1400	Fan cooled	400	70	8.3	4.0	1.1	3635	2153	920	1430	1723	689	4700
RCDC-16	1600	Fan cooled	450	70	11.0	5.5	1.1	3950	2360	1070	1640	1923	768	7100

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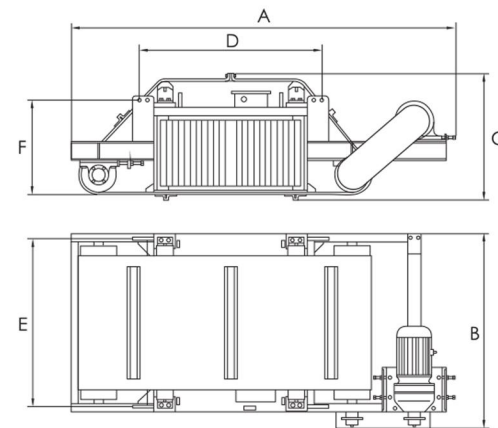
> RCDD Air-cooled Self-cleaning Electromagnetic Separator

Application

Suitable for heavy-dust working site.

Feature

- Compact structure, easy maintenance.
- Automatic rectification of departure of discharge belt.
- Low temperature rise.
- Continuous removal of tramp iron.



Technical Parameter

Model	Applicable Belt Width	Cooling Style	Rated Suspension Height	Magnetic Intensity at RSH	Excitation Power	Motor Power	Overall Dimension mm						Weight
	mm		mm	mT	kw	kw	A	B	C	D	E	F	kg
RCDD-5	500	Air cooled	150	70	1.8	1.5	2400	930	732	936	723	461	930
RCDD-6	650	Air cooled	200	70	2.5	1.5	2500	1054	732	1036	873	461	1400
RCDD-8	800	Air cooled	250	70	3.4	2.2	2770	1350	744	1142	1073	481	2100
RCDD-10	1000	Air cooled	300	70	4.4	3.0	3040	1550	822	1272	1273	504	2900
RCDD-12	1200	Air cooled	350	70	7.6	4.0	3275	1800	980	1320	1523	607	3900
RCDD-14	1400	Air cooled	400	70	9.1	4.0	3635	2000	920	1430	1723	692	5000
RCDD-16	1600	Air cooled	450	70	11.0	5.5	3950	2300	1070	1640	1923	788	7600

We reserve the right to change the above specification without prior notice.

> RCDE/F Oil-cooled Electromagnetic Separator

Application

Suitable for industries such as coal handling, power station, chemical, mining, and construction. It can withstand heavy dust, and corrosive environment.

Feature

- Quality cooling oil and optimized oil circuit design guarantees low temperature rise.
- Compact structure, light weight, low noise, simple operation, easy maintenance.



RCDE



RCDF

Technical Parameter

Model	Applicable Belt Width	Cooling Style	Rated Suspension Height	Magnetic Intensity at RSH	Excitation Power	Overall Dimension mm				Weight
	mm		mm	mT	kw	A	B	C	D	kg
RCDE-6	650	Oil cooled	200	70	—	926	900	370	870	720
RCDE-8	800	Oil cooled	250	70	—	1133	1050	460	960	1290
RCDE-10	1000	Oil cooled	300	70	—	1333	1250	520	1020	2130
RCDE-12	1200	Oil cooled	350	70	0.55	1640	1850	740	1160	2850
RCDE-14	1400	Oil cooled	400	70	0.55	1660	1860	830	1170	3200
RCDE-16	1600	Oil cooled	450	70	0.55	1700	1860	920	1280	4380

Model	Applicable Belt Width	Cooling Style	Rated Suspension Height	Magnetic Intensity at RSH	Excitation Power	Overall Dimension mm				Weight
	mm		mm	mT	kw	A	B	C	D	kg
RCDF-6	650	Oil cooled	200	70	—	2300	926	370	870	1300
RCDF-8	800	Oil cooled	250	70	—	2500	1133	460	960	1900
RCDF-10	1000	Oil cooled	300	70	—	2800	1333	520	1020	2800
RCDF-12	1200	Oil cooled	350	70	0.55	3250	2030	790	1210	4120
RCDF-14	1400	Oil cooled	400	70	0.55	3280	2240	840	1220	4860
RCDF-16	1600	Oil cooled	450	70	0.55	3600	2430	930	1390	6400

We reserve the right to change the above specification without prior notice.

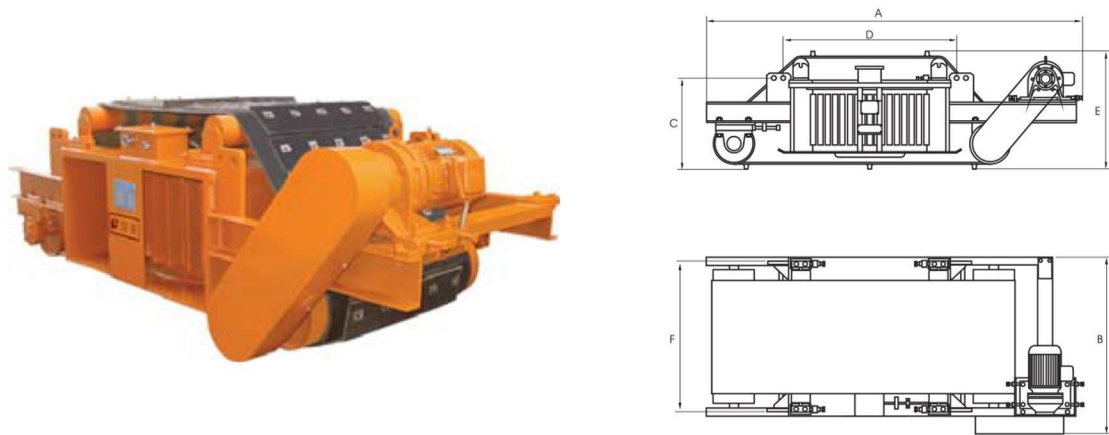
> RCDK Cladded Belt Electromagnetic Separator

Application

Suitable for concrete recycling, construction, demolition and waste processing.

Feature

- 1mm to 5mm stainless steel clad protect belt effectively.
- Safe and reliable insulation of the coils.
- Self-cleaning delivers convenient operation.
- Compact structure, easy maintenance, automatic rectification of departure of discharge belt, and long-term reliable operation.



Technical Parameter

Model	Applicable Belt Width	Cooling Style	Rated Suspension Height	Magnetic Intensity at RSH	Excitation Power	Duty Cycle	Overall Dimension mm						Weight
	mm		mm	mT	kw		A	B	C	D	E	F	kg
RCDK-5	500	Air cooled	150	≥70	1.8	1.5	2210	1045	645	945	673	486	1080
RCDK-6	650	Air cooled	200	≥70	2.5	1.5	2350	1113	702	1052	873	537	1570
RCDK-8	800	Air cooled	250	≥70	3.4	2.2	2460	1253	833	1110	1073	647	2100
RCDK-10	1000	Air cooled	300	≥70	4.4	3.0	3750	1593	945	1310	1273	760	3120
RCDK-12	1200	Air cooled	350	≥70	7.6	4.0	3080	1713	1050	1610	1523	840	4350
RCDK-14	1400	Air cooled	400	≥70	9.1	4.0	3450	1973	1165	1920	1723	947	5950

We reserve the right to change the above specification without prior notice.

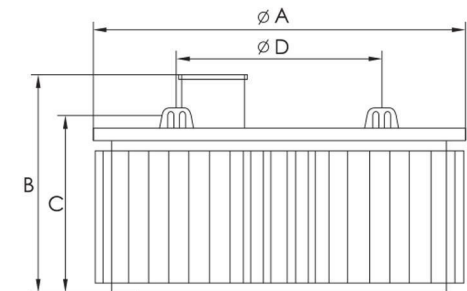
> PDC Manual-cleaning Electromagnetic Separator

Application

Applied in various working site especially for waste recycling industry.

Feature

- Small volume, light weight, compact structure
- Maintenance free
- Noise free, effective heat dissipation, low temperature rise
- local and remote control available



Technical Parameter

Model	Applicable Belt Width	Cooling Style	Rated Suspension Height	Magnetic Intensity at RSH	Excitation Power	Overall Dimension mm				Weight
	mm		mm	mT	kw	A	B	C	D	kg
PDC-5	500	Air cooled	150	70	1.8	810	431	353	610	500
PDC-6	650	Air cooled	200	70	2.5	905	519	441	700	830
PDC-8	800	Air cooled	250	70	3.4	1000	609	536	800	1300
PDC-10	1000	Air cooled	300	70	4.4	1080	701	628	850	1800
PDC-12	1200	Air cooled	350	70	7.6	1180	781	708	900	2500
PDC-14	1400	Air cooled	400	70	9.1	1300	841	768	1000	3150
PDC-16	1600	Air cooled	450	70	11.0	1350	933	860	1050	4100

We reserve the right to change the above specification without prior notice.

> LES Eddy Current Separator

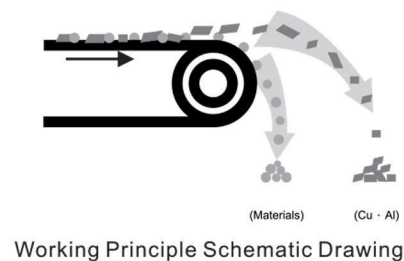
Feature

Eddy Current Separator (ECS) employs a conductive or repelling force between the magnetic field generated inside a unique rotor to throw or separate the non-ferrous metal out from other materials such as plastic, paper or other non-metallic material in the waste stream. The ECS has a wide application in recycling industry like aluminum recovery from crushed automobiles, refrigerators, scrap aluminum, etc. It is available in concentric and eccentric, and with feeding width ranges from 500mm to 2000mm.

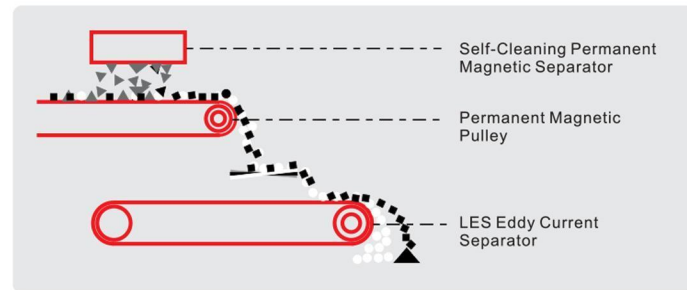


Feature

- LES eddy current separator is designed on the basic theory that introductive current generates whilst conductor is the high-frequency alternating magnetic field.
- When LES eddy current separator works, high-frequency alternating magnetic field generates on the surface of the magnetic rotor.
- When non-ferrous metal enter the high-frequency alternating magnetic field, eddy current occurs inside the non-ferrous metal.
- The direction of the magnetic field by eddy current is counter to that of the magnetic field by the high-frequency alternating magnetic field, which leads to rejecting force.
- The rejecting force will forward-throw non-ferrous metals off the high-frequency alternating magnetic field, so non-ferrous metals can be recycled effectively.

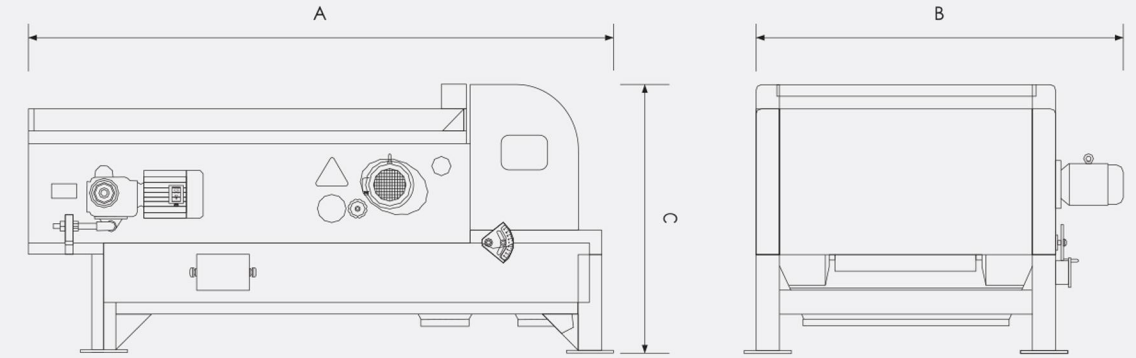


Working Principle Schematic Drawing



Working Process Schematic Drawing

> LES Eddy Current Separator



Main Technical Parameters

Model	Belt Width Adaptive	Motor Power	Handling Capacity	Outline Dimensions A×B×C	Weight
	mm	kw	m³/h	mm	kg
LES-60	650	4.0	6.0	2250×1750×1075	1100
LES-80	800	5.5	7.5	2360×1960×1110	1200
LES-100	1000	5.5	9.0	2470×2200×1140	1400
LES-120	1200	7.5	12.0	2560×2450×1180	1500
LES-150	1500	7.5	13.5	2650×2750×1215	1750
LES-180	1800	11.0	18.0	2740×3150×1245	1900
LES-200	2000	11.0	20.5	2850×3350×1270	2000

The magnetic intensity can be specially designed according to your requirements.



> LONGi Service

One-to-one Service

LONGi laboratory undertakes professional sample testing and reporting for a wide range of minerals and some of recycling industries. Based on test result and necessary site survey, experienced technical engineers will determine the most suitable solution for each individual application that will deliver utmost performance and economy to our customers.

Valuable Site Service

LONGi can provide experienced mechanical, electrical and magnetic engineers to assist with installation, commissioning, training, eliminating problems that can occur at the working site. Our offices are located with easy access to working sites to provide strategic spare parts and service.

Considerate Feedback Service

LONGi talks to the customers periodically to check operation of sold equipment, and assists to made necessary adjustment according to practical conditions to achieve utmost profit of the customer.